ENGR 111
Introduction to Engineering
Class #8

Job functions and the technology team

Same degree, very different days

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

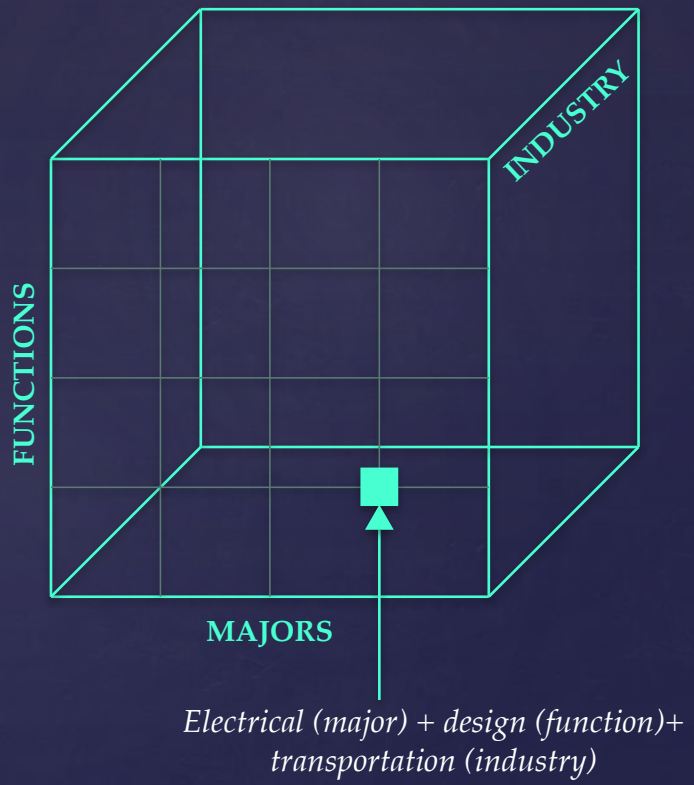
Three questions, not one

You have picked one axis. A major, or a shortlist of them. Choosing a major does not choose the work.

The second axis is what you do all day. Analysis, design, test, sales, management. Most students have not thought about this one yet.

The third is what industry you do it in. Transportation, energy, medical, construction, consumer products. Most have not thought about this one either.

Same major, different cell, different job. Two mechanical engineers, one designing medical devices and one testing bridges, have almost nothing in common day to day.



Which cell of that cube are you aiming at?

The ten job functions

An electrical engineer can be a design engineer, a test engineer, a sales engineer, etc.

Analysis — models the problem in mathematics before anyone builds anything.

Design — turns the concept into the drawings and specifications people build from.

Test — proves the thing meets its specifications, under the conditions it will see.

Development — sits between design and test, and builds the prototype.

Research — looks for new knowledge engineering can use. Usually a PhD.

Sales — the technical person between the company and the customer.

Management — line management up the organization, or one project end to end.

Consulting — sells expertise to clients by contract, sometimes for a day.

Teaching — teaching, research and service.

Entrepreneur — starts the company.

Which of these ten had you not thought of as an engineering job?

One product, many different jobs

An electric bike, and one example task for each of seven functions.

Analysis. Works out how much motor power the hill actually needs, before anyone buys a motor.

Design. Picks the motor, the gear ratio, the cell chemistry, and draws the frame.

Development. Builds three of them and rides them.

Test. Rides it in the rain, in the cold, and through five hundred charge cycles, and checks the range number still holds.

Sales. Explains to a fleet buyer why this one and not the cheaper one.

Management. Manages the project including people, schedule, budget, etc.

Consulting. Brought in for two weeks because nobody in-house has done battery safety before.





What would a research engineer do in this field?

Where engineers actually work

From a national survey of people who hold engineering degrees.

Where	Share
Business and industry	64.5%
Educational institutions	18.5%
State and local government	6.2%
Self-employed	6.2%
Federal government	4.6%

The top row splits in two. Those 64.5 percent are manufacturing, which converts raw materials into products, and non-manufacturing, which delivers services.

Two of the rows are government. State and local plus federal is 10.8 percent — public works, transport, defense, and the agencies that write the rules everybody else designs to.

What the table counts. People who hold an engineering degree, not people whose job title says engineer. That is part of why the education share looks so high.

Landis table 2.8, p.72. Not the same population as the BLS occupation counts.

The technology team

Six roles, six different routes in, and only one of them is an engineering degree.

Role	What they do	Usual route in
Technician	Assembly, test, troubleshooting, maintenance, calibration, drafting	Associate degree in engineering technology
Technologist	Modifies existing designs, writes procedures	Bachelor's in engineering technology
Engineer	Defines the problem, designs, analyzes, specifies, signs off	BS, MS, or PhD in Engineering
Scientist	Looks for new knowledge	Usually a PhD
Tradesperson	Welding, machining, electrical, plumbing, fabrication	Apprenticeship, sometimes an associate degree
Manager	Schedule, budget, people	Varies. Often engineering plus an MBA

Which of these roles have you already held or worked alongside?

Engineering vs engineering technology

ABET's summary: theory and design against application and implementation.

	ENGINEERING · EAC	ENGINEERING TECHNOLOGY · ETAC
Mathematics	At or above introductory calculus.	Bachelor's: calculus, or math above algebra and trigonometry. Associate: algebra and trigonometry.
Math and science credits	At least 30 semester credits, with experimental experience.	No credit-hour floor. Natural science with labs, suited to the discipline.
Technical content	At least 45 semester credits of engineering topics.	One third to two thirds of all credits, discipline-specific.
Design	A culminating major design experience, with standards and multiple constraints.	Design considerations in the technical core. Bachelor's also needs a capstone.
General education	A broad education component.	Technical, professional and general education combined.
Graduates are called	Engineers.	Technologists, from a four-year program.

ABET EAC and ETAC accreditation criteria, 2026-27 cycle.

BSEE vs BSEET at Oregon Tech

The two degrees run side by side at Oregon Tech. Math is mostly where they separate.

ENGINEERING ONLY	BOTH DEGREES	TECHNOLOGY ONLY
Sequences and Series	Differential Calculus	College Algebra
Linear Algebra	Integral Calculus	Trigonometry
Mathematical Statistics	Vector Calculus	Statistical Methods
	Applied Differential Equations	

Both go through differential equations. The middle column is the same for the two degrees at this school. Engineering technology is not a way around calculus.

They separate at both ends. Technology starts lower, at algebra and trigonometry. Engineering carries on into linear algebra and mathematical statistics.

This is one program, not the national rule. ABET's floor for an engineering technology bachelor's is lower than what you see here. Read the actual course list.

Clackamas has one of each. An AS with an engineering emphasis is built to transfer into a four-year engineering degree (like BSEE). The AAS in Electronics Engineering Technology transfers into Oregon Tech's four-year technology degree (like BSEET) with junior standing.

Which of those courses have you already taken?

What the graduates actually do

First job titles reported by graduates of Oregon Tech's BSEE and BSEET programs

ENGINEERING (EE)	ENGINEERING TECHNOLOGY (EET)
Electrical engineer	Design engineer
Computer engineer	Hardware design engineer
Substation engineer	Technical marketing engineer
Firmware engineer	Test process engineer
Project engineer	Project engineer
Electrical equipment technician	Engineering technician, test technician

The two lists overlap. Design engineer, project engineer and test roles turn up on both sides, and both programs feed the same employers: Intel, Tektronix, AMD, Garmin.

Four year against four year. In a 2017 survey of these two Oregon Tech programs, the technology graduates started at a slightly higher median than the engineering graduates. Most of them arrived with an associate degree and years on the job.

The national figures do not split the two. Median pay in May 2024 was $111,910 for electrical engineers and $77,180 for engineering technologists and technicians — but that second occupation counts four-year technologists together with associate-degree technicians, and lists an associate as the usual way in.

Data: Oregon Tech program outcomes, 2017; BLS Occupational Outlook Handbook, May 2024.

Which of these fits you?

The job you think you want if often different than what you would be good at.

Each function demands something different. Sales needs people skills and a lot of travel. Analysis needs patience with mathematics and long stretches alone with a problem.

What appeals to you and what fits you are two different lists. Working out what a job demands comes before deciding whether it is you.

Day-to-day will vary drastically between different industries and employer. The job function and title actually tells you very little about what the day-to-day is like.

It is easier to change function than to change discipline. Design to test does not need another degree. Civil to chemical does.

Plenty of engineers end up in a function they never planned on — and often enjoy it.





